

Joshua Cox

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Education

University of Illinois Urbana-Champaign (UIUC)

M.S. in Mechanical Engineering

GPA: 3.83 / 4.00

December 2025

B.S. in Mechanical Engineering with Honors | Minor in Computer Science

GPA: 3.61 / 4.00

December 2022

Research Experience | Human-Centered Autonomy Lab

Lunar Autonomy Challenge

November 2024 – December 2025

- Founded and led a student robotics team to compete in NASA simulation challenge, creating an autonomous lunar rover
- Engineered the perception to mapping pipeline through utilizing ROS, visual inertial odometry solutions, and a custom computer vision algorithm to convert stereo pairs to accurate point clouds
- Processed elevation data by developing visualization tools, trying elevation filters, and testing grid painting techniques
- Implemented navigation strategy using both a high-level waypoint generator and a low-level obstacle avoidance planner
- Developed 2D exploration simulator from scratch with Python and PyGame to test team's navigation algorithm
- Created modular Dockerfile to incorporate competition and solution requirements enabling swift software development

Wayfinding with Handle Haptics

February 2024 – December 2025

- Designed, created CAD for, and 3D-printed parts for a novel haptic handle attached to a wayfinding robot in order to communicate signals and provide feedback to guided users
- Rapidly prototyped designs to test mounting strategies, vibration magnitude, and user comfort

Reinforcement Learning Autonomous Agent

August 2023 – December 2025

- Utilized CARLA simulator for testing autonomous control and gathering data on a mobile robot (self-driving car)
 - Implemented reinforcement learning-based controller for an autonomous agent to successfully navigate a circular track
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Work Experience

Controls Lab Teaching Assistant | UIUC | Urbana, IL

January 2023 – December 2025

- Led lab sections for courses in control systems, signals, and embedded systems, delivering instruction and guidance
- Mentored students through end-to-end development of an embedded robotic vehicle, including PCB soldering, bare-metal C firmware, sensor integration, and closed-loop control implementation
- Diagnosed and resolved electrical, mechanical, and software issues in experimental setups and data acquisition systems using oscilloscopes, multimeters, and software debugging tools
- Supported new teaching assistants, contributed to lab curriculum, and improved documentation and lab infrastructure

Embedded Systems Engineer Intern | Engineering System Design Lab | Urbana, IL

May 2024 – August 2024

- Completed hardware and software tasks to develop a new testing setup for a hinge-integrated multifunctional structures for attitude control (MSAC) scheme for satellites utilizing piezoelectric strain actuators
 - Designed and implemented bare-metal C++ firmware for STM32 Nucleo-H755ZI-Q, delivering advanced PWM capabilities, reliable SD card logging, and synchronized communication with multiple accelerometers
 - Tested types of signal filters to produce sin signals from period-changing PWM signals as inputs for the actuators
 - Built MATLAB tools to analyze frequency response, ensure accurate signal generation, and verify reliable data logging
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Project Experience

Hand-Gesture Controlled Robotic Arm

August 2023 – December 2023

- Developed code to recognize hand poses using MediaPipe's model trained on the HaGRID dataset
- Controlled UR3 robot arm to draw using a pen attachment and to pick and place toy blocks based on user hand poses
- Designed and 3D-printed a pen holding attachment mounted to UR3 robot arm, enabling it to draw on paper

Spaceshot Rocketry Team

August 2021 – December 2022

- Assembled active roll control system composed of rack-and-pinion servos and tested its clearances required during flight
 - Developed RTOS-based embedded firmware to log sensor data and control servos using a real-time feedback algorithm
 - Validated new avionics stack performance onboard test rocket, which successfully guided rocket to 10,000 feet, deployed control flaps when necessary, and recorded flight telemetry for future analysis
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Skills

Programming Languages: Python, C/C++, Java, C#, Microsoft VBA, Bash Scripting

Software & Tools: MATLAB, Simulink, Creo Parametric, PyTorch, Docker, Git, ROS, STM Studio, Godot, PyGame, Fusion, Solidworks, Unity, CARLA Simulation, Gazebo Simulation, LaTeX, OpenCV

Hardware & Microcontrollers: STM32, TI C2000, Stretch3, TurtleBot, UR3 robot arms, ESP32, IMUs, stereo cameras

Other: 3D printing, design, rapid prototyping, oscilloscope use, technical writing, signal filtering, cleanroom operation